see attached notes

next power of 2

```
dec     rax
```
$rax = rax_0 - 1$

```
mov     rdx,rax
shr     rdx,0x1
or      rax,rdx
```
$rax = rax_0 - 1$

$rdx = (rax_0 - 1) >> 1$

$rax = (rax_0 - 1) \vee (rax_0 - 1) >> 1$

```
mov     rdx,rax
shr     rdx,0x2
or      rax,rdx
```
is

$x \leftarrow x \vee (x >> 2^k)$

$k = 0,1,2,3,4,5$

```
mov     rdx,rax
shr     rdx,0x4
or      rax,rdx
```

```
mov     rdx,rax
shr     rdx,0x8
or      rax,rdx
```
$x_* = \min\{2^n\}$

$\ni 2^n \geq x_0$

```
mov     rdx,rax
shr     rdx,0x10
or      rax,rdx
```
is

"round up to next power of 2"

```
mov     rdx,rax
shr     rdx,0x20
or      rax,rdx

inc     rax
```

```
dec rax
```

rax --

1111 .... 1111 → 1111 ... 1110

```
mov rdx, rax
shr rdx, 0x1
or  rax, rdx
```

rdx := rax
/2 → rdx = 0111 1111 ... 1111
or 1111 1111 ... 1110
rax = 1111 ... 1111

```
mov rdx, rax
shr rdx, 0x2
or  rax, rdx
```

rdx := 0111 ... 1111
or: 1111 1111 ... 1111
rax = 1111 ... 1111

```
mov rdx, rax
shr rdx, 0x4
or  rax, rdx
```

shr
rdx = 0000 1111 1111
or 1111 1111 ... 1111
rax 1111 ... 1111

```
mov rdx, rax
shr rdx, 0x8
or  rax, rdx
```

0000 0000 1111 ... 1111
or f f f ... f
~~ffff ffff~~

shr: ~~0000 ffff~~

```
mov rdx, rax
shr rdx, 0x10
or  rax, rdx
```

```
mov rdx, rax
shr rdx, 0x20
or  rax, rdx
```

0000 ... 0000
1111 ... 1111
r1
||
0...0

```
inc rax
```

(3) → 2

00 ... 0011 → 0 ... 0010
+
0001
or
0011 3

0000
0011
||

1000
+
0111
0001

8

3+1=4

0000 ... 0000

1000 ... 0000
or 0100 ... 0000
1100 ... 0000

11 00 00 ... 0000
or 1111 0000 ... 0000

1111 0000 ... 0000
1111
16 16
1 ... 1 0 ... 0

1111 ...
111 ...

1111 ... 1111
32
1111 ... 1111
or 0000 ... 0000
~~0xffff ffff~~ + 1
||
0

highest value 1-5 out the rest

ex) 0000 10 ... 0

111 ... 1

(a = rax
min{2^n} st 2^n > a)

0 11 1111 - 1 ⇒ 0 ... 011 1110 → 011 1111 ≠ 1

000 ... 01 00 0000

1111 1111 - 1 ⇒ 1111 1110

~~no fields need go~~